Ruvarashe Shalom Madzime

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RESEARCH INTERESTS

Formal non-monotonic reasoning, in particular the KLM framework, and its use for giving learned rule sets a logical reading. Moving toward neurosymbolic AI, focused on how logical structure can be embedded into neural networks during training and how networks might learn symbolic representations directly from data.

EDUCATION

MSc Computer Science **February 2025 to November 2026 (expected)**

University of the Western Cape and Centre for AI Research (CAIR), Cape Town, South Africa

Thesis: *Toward a Defeasible Semantics for Symbolic Classifiers*. Supervisors: Prof. Thomas Meyer, Prof. Louise Leenen. Area: knowledge representation, defeasible reasoning, interpretable machine learning.

BSc (Hons) Computer Science, cum laude (85%) **January 2024 to November 2024**

University of the Western Cape, Cape Town, South Africa

Modules: artificial intelligence, applied data science, Internet of Things, cybersecurity, big data engineering.

BSc Computer Science, cum laude (84%) **March 2021 to November 2023**

University of the Western Cape, Cape Town, South Africa

Key modules: artificial intelligence, mathematics, statistics.

PUBLICATIONS

[1] Madzime, R., and Meyer, T. (2026). Toward Defeasible Machine Learning: When Learned Rule Sets Become Defeasible Theories. In *Proceedings of the 24th International Workshop on Nonmonotonic Reasoning (NMR 2026)*, Federated Logic Conference (FLoC), Lisbon, Portugal.

[2] Karge, J., Ferguson, R., Grimaldi, D., Haldimann, J., Madzime, R., and Meyer, T. (2026). First Steps Towards Human-AI Ranking Aggregation. In *Proceedings of the Joint Workshop on Statistics and Knowledge Integration for Logic, Learning, Ethical Decisions, and LLMs (SKILLED-LLMs 2026)*, Federated Logic Conference (FLoC), Lisbon, Portugal.

[3] Madzime, R. S., Meyer, T., and Leenen, L. (2025). An Override-Aware Classifier for Transparent AI. In *Proceedings of the 6th Southern African Conference on Artificial Intelligence Research (SACAIR 2025)*, Cape Town, South Africa.

[4] Madzime, R., and Nyirenda, C. (2024). Enhanced Electronic Health Record Text Summarisation Using Large Language Models. In *Proceedings of the 5th Southern African Conference on Artificial Intelligence Research (SACAIR 2024)*, Bloemfontein, South Africa.

RESEARCH AND TEACHING EXPERIENCE

Research Assistant **March 2026 to present**

SARChI Chair in Knowledge Representation (Prof. Louise Leenen), University of the Western Cape

Contribute to research in knowledge representation as part of the NRF SARChI programme. Provide teaching support for undergraduate and postgraduate Computer Science modules. Co-supervise Honours research projects, giving feedback on methodology, writing, and experimental design.

Graduate Lecturing Assistant **February 2025 to November 2025**

Department of Computer Science, University of the Western Cape

Ran tutorials and one-on-one consultations for Computer Science modules for over 50 students. Graded assignments and exams, and supported lecturers with coursework design and student queries.

Graduate Research Assistant **January 2024 to November 2024**

Department of Computer Science, University of the Western Cape

Built an LLM-based summarisation system for Electronic Health Records, published at SACAIR 2024. Designed the full pipeline from dataset analysis through model evaluation using Python and standard ML tooling. Contributed to a funding proposal by reviewing existing literature and identifying gaps in EHR summarisation.

Python Software Developer **October 2023 to March 2024**

Apparel Technology Inc., Cape Town, South Africa

Built a Python and Arduino monitoring system that tracked factory machine uptime and productivity in real time. Created a data visualisation dashboard with Python Dash for the operations team.

HONOURS AND AWARDS

- UWC Mastercard Foundation Scholarship, 2025 to 2026.
- Top Computer Science Student, University of the Western Cape, 2023.
- KnowBe4 Women in Cybersecurity Scholarship, 2023.
- Dean's Merit Award, University of the Western Cape, 2021, 2022, 2023, and 2024.

SKILLS

Programming and tools. Python, Java, SQL, LaTeX, Git.

Symbolic and formal methods. Answer Set Programming (Clingo), SMT solvers (Z3), defeasible reasoning, knowledge representation.

Machine learning. PyTorch, scikit-learn, Hugging Face, SHAP, natural language processing, large language models, retrieval-augmented generation.

Other. Technical writing, data visualisation, cloud deployment.